

Problem 3 in Moslehian's complemented subspace survey is answered

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Status. `literature_already_answered`. This packet records a later explicit counterexample to one of the open problems listed in Mohammad Sal Moslehian's survey [1].

Source question. In Section 12, "Some interesting problems," Problem 3 asks:

Is every complemented vector subspace of $C(S)$
isomorphic to some $C(S_1)$?

The problem appears on PDF page 10 of [1]. In modern terminology, this asks whether the class of $C(K)$ -spaces is closed under complemented subspaces.

Supporting answer. Plebanek and Salguero Alarcon [2] explicitly formulate the complemented subspace problem for C -spaces as Problem 1.1 and then give a negative answer. Their abstract states that they construct a compact space L and a 1-complemented subspace of $C(L)$ which is not isomorphic to a space of continuous functions. Their Theorem 1.1 gives the precise construction: there are compacta K, L , a continuous surjection $\theta : L \rightarrow K$, and a closed subspace $X \subset C(L)$ such that

$$C(L) \simeq \theta^\circ[C(K)] \oplus X,$$

both summands are 1-complemented in $C(L)$, and X is not a C -space.

Identification. This is exactly a negative answer to Moslehian's Problem 3. The supporting paper does not merely imply the answer indirectly: it names the complemented subspace problem for C -spaces and presents its main theorem as a counterexample.

Scope limitations. The source survey lists seven broad problems. This packet clears only Problem 3. It does not claim answers to the general complemented subspace classification problem, the injective-space question, the von Neumann algebra question, or the prime-space question.

Search evidence. The run's cheap indexes were searched for arXiv:0501048, arXiv:2111.13860, the source title, and phrases around complemented subspaces of $C(K)$ -spaces. No prior packet for this source/supporting-pair answer was present. A web search for the same phrases located arXiv:2111.13860 as the direct later counterexample.

References

- [1] M. S. Moslehian, *A Survey on the Complemented Subspace Problem*, arXiv:math/0501048, 2005. <https://arxiv.org/abs/math/0501048>
- [2] G. Plebanek and A. Salguero Alarcon, *The complemented subspace problem for $C(K)$ -spaces: A counterexample*, arXiv:2111.13860, 2021. <https://arxiv.org/abs/2111.13860>